

CAA 25-26

Lab #2

29-10-2025

Introduction

In this lab, you are going to code review a simple Python app. This app allows patients to input their symptoms so that a doctor can verify them and give them a prescription. It also allows doctors to create prescriptions for a patient. All the records are signed with the platform's private key so that doctors know they read authentic symptoms and pharmacies know they read valid prescriptions.

Your goal is to review **the cryptography** used in the app (and only the cryptography!). Hence, you do **not** have to look at safe logins, secure coding problems that do not affect crypto, databases, ...

If you **use an AI** for helping you, be careful at what is output. Additionally, **clearly state how you used AI** as asked by the school regulation: heig.ch/ia

1 Tasks

Write a **report** in which for each error:

1. You show which lines of code is problematic
2. You explain in technical terms why it is problematic and what are the risks. In particular, explain what an attacker can do and what can go wrong.
3. You explain in high level terms (for your boss that doesn't know crypto) why it should be fixed.
4. You give a fix.

Sort the errors by **criticality** in your report (starting with the most critical one).

Additionally, provide an **application** in which all problems have been fixed.